

Lab-7: Packet-level Firewalls – Access Control Lists (ACLs)

Objectives

- Use of Access Control Lists (ACLs) defined in Routers to control access in a network.
- Difference between Standard and Extended ACLs.
- Configuring and applying Standard and Extended ACLs in Cisco routers.

Part-1: Access Control Lists (ACLs)

The **Access Control List (ACL)** is a collection of security rules or policies that allows or denies packets after looking at the packet headers and other attributes. Each permit or deny statement in the ACL is referred to as an **access control entry (ACE)**. These **ACEs** can classify packets by inspecting Layer 2 through Layer 4 headers for a number of parameters, including the following:

- Layer 2 header information such as Ether Types
- Layer 3 header information such as ICMP, TCP, or UDP
- Layer 3 header information such as source and destination IP addresses
- Layer 4 header information such as source and destination TCP or UDP ports

After an ACL has been properly configured, you can apply it to an interface to filter traffic. The security appliance can filter packets in both the inbound and outbound direction on an interface. When an inbound ACL is applied to an interface, the security appliance analyzes packets against the ACEs after receiving them. If a packet is permitted by the ACL, the firewall continues to process the packet and eventually passes the packet to the defined interface.

Two types of Access Control Lists are,

1. Standard Access Control Lists (ACLs):

Standard access control lists (ACLs) are router configuration scripts that control whether a **router permits or denies packets based on the source address only**.

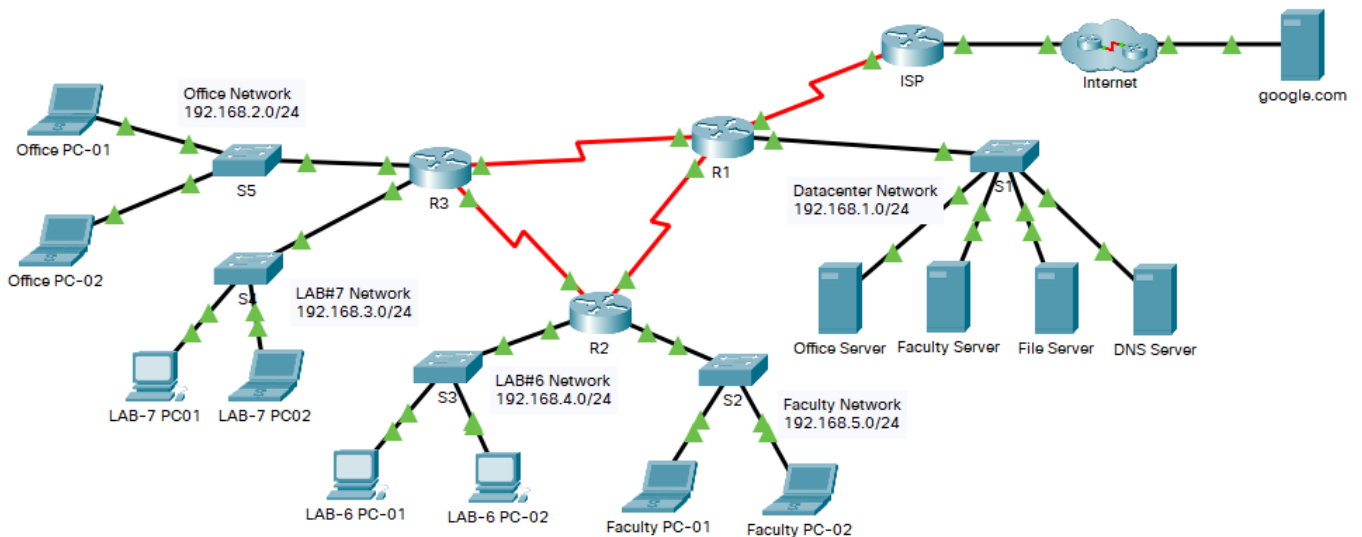
Standard ACL defined using any number from 1 to 99.

2. Extended Access Control Lists (ACLs):

Extended access control lists (ACLs) are extremely powerful. They offer a much greater degree of control than standard ACLs as to the types of traffic that can be filtered, as well as where the traffic originated and where it is going. Extended ACLs can filter traffic in many different ways. **Extended ACLs can filter on source IP addresses, source ports, destination IP addresses, destination ports, as well as various protocols and services.**

Extended ACL defined using any number from 100 to 199.

Part-2: Access Control Lists (ACLs) Implementation



Part-2(A): Standard Type Access Control Lists (ACLs)

Create an ACL using the number on R1 (Why not on R2 or R3?) with a statement that denies access to the Datacenter Network (192.168.1.0/24) from the LAB#7 Network (192.168.3.0/24).

```
R1 (config)# access-list 1 deny 192.168.3.0 0.0.0.255
```

By default, an access list denies all traffic that does not match a rule. To permit all other traffic, configure the following statement:

```
R1 (config)# access-list 1 permit any
```

For the ACL to actually filter traffic, it must be applied to some router operation. Apply the ACL by placing it for outbound traffic on the Gigabit Ethernet 0/0 interface.

```
R1 (config)# interface GigabitEthernet0/0
```

```
R1 (config-if)# ip access-group 1 out
```

```
R1 (config-if)# exit
```

Prevent (Deny) LAB#7 PC01 to Access Datacenter Network.

```
R1(config)#access-list 2 deny host 192.168.3.10
```

```
R1(config)#access-list 2 permit any
```

```
R1(config)#int gigabitEthernet 0/0
```

```
R1(config-if)#ip access-group 2 out
```

```
R1(config-if)#exit
```

Prevent all Private IP Address to Access Internet (Please follow the order)

```
ISP(config)#access-list 10 deny 10.0.0.0 0.255.255.255
ISP(config)#access-list 10 deny 172.16.0.0 0.15.255.255
ISP(config)#access-list 10 deny 192.168.0.0 0.0.255.255
ISP(config)#access-list 10 permit any
```

```
ISP(config)#int serial 0/0/0
ISP(config-if)#ip access-group 10 in
ISP(config-if)#exit
```

Part-2(A): Extended Type Access Control Lists (ACLs)

LAB#6 network is not allowed to access Office Server using http/https/browsing, but All other Traffic is Permitted for Lab#6 Network.

```
R2(config)#access-list 100 deny tcp 192.168.4.0 0.0.0.255 host 192.168.1.11 eq www
R2(config)#access-list 100 permit ip any any
```

```
R2(config)#int gigabitEthernet 0/1
R2(config-if)#ip access-group 100 in
R2(config-if)#exit
```

Office Server will be allowed to access only by Office Network, but all other networks will be denied. All networks will be allowed to access any other Datacenter Servers.

```
R1(config)#access-list 101 permit ip 192.168.2.0 0.0.0.255 host 192.168.1.11
R1(config)#access-list 101 deny ip any host 192.168.1.11
R1(config)#access-list 101 permit ip any any
```

```
R1(config)#int gigabitEthernet 0/0
R1(config-if)#ip access-group 101 out
R1(config-if)#exit
```

LAB-7 PC01 cannot ping File server but only access by http, All other Traffic is Permitted for Data Center Network.

Answer-01: (in Router R1)

```
R1(config)#access-list 102 deny icmp host 192.168.3.10 host 192.168.1.14 echo-reply
R1(config)#access-list 102 permit tcp host 192.168.3.10 host 192.168.1.14 eq www
R1(config)#access-list 102 deny ip host 192.168.3.10 host 192.168.1.14
R1(config)#access-list 102 permit ip any any
```

```
R1(config)#int gigabitEthernet 0/0
R1(config-if)#ip access-group 102 out
R1(config-if)#
```

Answer-02: (in Router R3)

```
R1(config)#access-list 103 deny icmp host 192.168.3.10 host 192.168.1.14 echo-reply
R1(config)#access-list 103 permit tcp host 192.168.3.10 host 192.168.1.14 eq www
R1(config)#access-list 103 deny ip host 192.168.3.10 host 192.168.1.14
R1(config)#access-list 103 permit ip any any
```

```
R1(config)#int gigabitEthernet 0/0
R1(config-if)#ip access-group 103 in
R1(config-if)#
```